Challenge: PULSE — Smart Hospital Infrastructure via Bluetooth PDoA and Auracast

Jorg Wieme ✉
jorg.wieme@ugent.be
IDLab, Ghent University - imec
Ghent, Belgium

Adelina Giurea
adelina.giurea@ugent.be
IDLab, Ghent University - imec
Ghent, Belgium

Eli De Poorter
eli.depoorter@ugent.be
IDLab, Ghent University - imec
Ghent, Belgium

Jeroen Hoebeke
jeroen.hoebeke@ugent.be
IDLab, Ghent University - imec
Ghent, Belgium

Abstract

Modern healthcare systems require indoor localization and information dissemination for patient safety and operational efficiency, yet current solutions treat tracking and communication as separate problems and scale poorly. This paper presents the vision of Phase-based Unified Localization, Streaming and Emergency (PULSE), a Bluetooth-derived framework that combines tracking and communication on one synchronized infrastructure, replacing connection-oriented Bluetooth Channel Sounding with a one-way Phase Difference of Arrival (PDoA) scheme in which the tag's unknown initial phase cancels across anchor pairs. This shifts all coherence requirements from the wearable to the mains-powered backbone: tags remain low-power broadcast devices, while anchors act as phase-coherent receivers. The same infrastructure carries Auracast audio, extended with Virtual Broadcast Isochronous Group (vBIG) to support audio dissemination and a bidirectional emergency uplink within one network-wide schedule. We analyze the feasibility boundaries of this design, covering backbone synchronization, anchor coherence, multipath, and capacity, and outline a roadmap toward energy-efficient, next-generation care delivery.

CCS Concepts

• **Computer systems organization** → *Architectures*; • **Applied computing** → **Health care information systems**; • **Networks** → **Wireless personal area networks**.

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1 Introduction

Healthcare systems face increasing pressure from aging populations, growing care demands, and personnel shortages. Hospitals and other facilities must improve safety, accessibility, and efficiency while keeping complexity and energy consumption low.

Patient tracking and information dissemination are crucial, yet existing systems treat them as separate functions, resulting in fragmented architectures that cannot provide closed-loop control. Existing positioning technologies each fall short for this combined role: Wi-Fi is energy intensive, Bluetooth Low Energy (BLE) Received Signal Strength Indicator (RSSI) lacks sub-meter accuracy, and Ultra-wideband (UWB) requires additional hardware.

PULSE instead combines ranging, audio dissemination, and emergency uplink on a single Bluetooth radio. Rather than comparing localization technologies alone, PULSE replaces the fragmented stack of UWB localization, Auracast audio, and a separate uplink with a unified infrastructure. Offloading synchronization to a mains-powered backbone keeps wearables simple and low power.

Building on this, PULSE (i) replaces connection-based Channel Sounding (CS) with one-way PDoA across synchronized anchors, (ii) reuses the Auracast and vBIG [3] broadcast structure as a single network-wide schedule that also carries an emergency uplink, and (iii) maps each component of the vision to the principal challenges that determine its feasibility. We present PULSE as a vision rather than a finished system.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the PULSE architecture, while Section 4 analyzes its challenges and feasibility constraints. Finally, Section 5 outlines a validation roadmap.

2 Related Work

A representative system [2] combines passive UHF RFID for localization with 6LoWPAN telemetry, demonstrating continuous patient tracking. However, it relies on heterogeneous stacks and operates in a unidirectional manner, aggregating data centrally without enabling feedback.

Indoor localization has been widely studied, emphasizing trade-offs between accuracy, cost, and scalability [9]. BLE RSSI-based approaches can achieve sub-meter accuracy with fingerprinting in controlled healthcare settings [8], but remain sensitive to interference and show reduced performance in real deployments [10].

Angle-of-arrival systems based on Bluetooth Direction Finding track connectionless tags at low power, but require calibrated antenna arrays at every anchor and address localization only. More recently, the Bluetooth standard introduced CS. Early studies highlight the potential of CS in complex environments [11] and its applicability to localization [6]. To overcome the scalability limits of connection-oriented CS, [5] proposes a connectionless architecture based on Periodic Advertising with Responses (PAwR). While this approach improves scalability, it still relies on bidirectional communication within response slots. None of these technologies also disseminate information, leaving tracking and communication as separate stacks.

Bluetooth introduced Auracast for audio broadcasting, but its adoption in assistive listening is constrained by end-to-end latency: the broadcast is heard alongside the direct sound, and the delay is perceived as an echo [4]. A key limitation of Auracast is its unidirectional design. Therefore, [3] suggested the vBIG concept, enabling bidirectional communication for Auracast. vBIG remains focused on audio systems, and the use of Auracast as part of a closed-loop dissemination framework has not yet been explored.

3 PULSE Architecture

PULSE couples a connectionless PDoA tracking layer with an Auracast and vBIG broadcasting layer over a shared synchronized backbone. Figure 1 illustrates PULSE in a healthcare setting. All anchors connect to a central unit via high-speed links and operate as passive entities, forwarding raw phase and reception data. The central unit, ranging from a rule-based controller to a Digital Twin, maintains a spatial map to track devices, detect emergencies, and coordinate responses. For instance, a detected fall can alert nearby staff, closing the loop between sensing and actuation.

3.1 Tracking Layer

3.1.1 Bluetooth Channel Sounding. Introduced in Bluetooth Core Specification 6.0, CS, and particularly Phase-Based Ranging (PBR), improves ranging accuracy over RSSI while reducing Direction Finding hardware complexity. However, standard CS requires a persistent connection to schedule measurements and exchange data. Inline Phase Correction Term (PCT) Transfer (Core Specification 6.3) reduces this overhead but remains one-to-one; moreover, commercial controllers support only tens of concurrent connections. Maintaining many wearable-to-anchor connections increases overhead, energy use, and collisions, motivating the connectionless approach below.

3.1.2 Connectionless PDoA. In PULSE, the mobile tag periodically transmits carrier tones that are passively received by surrounding anchors, mirroring UWB Time Difference of Arrival (TDoA) systems. However, the limited bandwidth of Bluetooth channels makes the nanosecond timestamping required for TDoA impractical, motivating a phase-based alternative.

Consider a tag at unknown coordinate (x, y) and N synchronized anchors at known (x_i, y_i). For a tone at frequency f with initial phase ϕ_0, the phase at anchor i is:

$$\hat{\phi}_i^f = \phi_0 - 2\pi f \frac{d_i}{c} + \phi_{\mathrm{LO},i}^f + 2\pi f \tau_i \pmod{2\pi}, \tag{1}$$

where d_i is the tag-anchor distance, $\phi_{\mathrm{LO},i}^f$ the anchor's local oscillator (LO) phase on tone f, and τ_i its residual timing error relative to the backbone reference. The spatial phase difference of a synchronized pair Rx_1, Rx_2 cancels the tag term ϕ_0, since both observe the same transmission:

$$\hat{\phi}_{1,2}^f = \frac{2\pi f}{c}(d_2 - d_1) + \Delta\phi_{\mathrm{LO}}^f + 2\pi f \tau_{1,2} \pmod{2\pi}. \tag{2}$$

Notably, this cancellation holds per tone: the tag does not need to maintain phase coherence across frequency hops. The wearable therefore remains a simple broadcaster. Stepping the frequency by a constant Δf (1 MHz) resolves the modulo-2π ambiguity, and with $d_{1,2} = d_2 - d_1$:

$$d_{1,2} = \frac{c}{2\pi\Delta f}\Delta\hat{\phi}_{1,2}. \tag{3}$$

Each distance difference $d_{i,j}$ defines a hyperbola; combining measurements from at least three anchors yields two independent intersecting hyperbolas, uniquely resolving the tag location (x, y). The residual infrastructure terms $\Delta\phi_{\mathrm{LO}}^f$ and $\tau_{1,2}$, together with multipath, do not cancel and govern feasibility, Section 4 analyzes them.

3.2 Broadcasting Layer

3.2.1 Auracast. Auracast enables one-to-many audio broadcasting on top of BLE. Receivers synchronize once to a broadcaster's hopping pattern and timing, after which they periodically wake, receive, and return to sleep without a connection. A single broadcaster can expose multiple streams, supporting separation across use cases. In large environments, multiple broadcasters are required for coverage and must transmit in a synchronized manner, which the wired backbone of PULSE provides.

3.2.2 vBIG. To overcome the unidirectional limitation of standard Auracast, PULSE adapts vBIGs [3] so the same structure carries both communication and audio, rather than the bidirectional audio of the original design. Each vBIG event is partitioned into four logical channel categories, as illustrated in Figure 2: downlink audio streams for continuous dissemination, scheduled uplink channels carrying PDoA tone sweeps and lightweight data, emergency channels for unscheduled distress signaling, and handover channels to request future uplink access. The event structure thereby doubles as the network-wide schedule, so no separate coordination protocol is needed and tags know exactly when to transmit or sleep. Its capacity and contention implications are examined in Section 4.

4 Challenges and Feasibility

PULSE is a vision, and its viability rests on several open challenges. Table 1 maps each component to its principal challenges, discussed in turn below.

4.1 One-Way PDoA

The appeal of the one-way scheme is that the tag's phase cancels in the spatial difference, but that same cancellation leaves several infrastructure-side terms in Eq. (2) untouched. Two of these terms, together with multipath, determine how accurately the anchors can recover a position, and they control the design of both the anchors and the backbone.

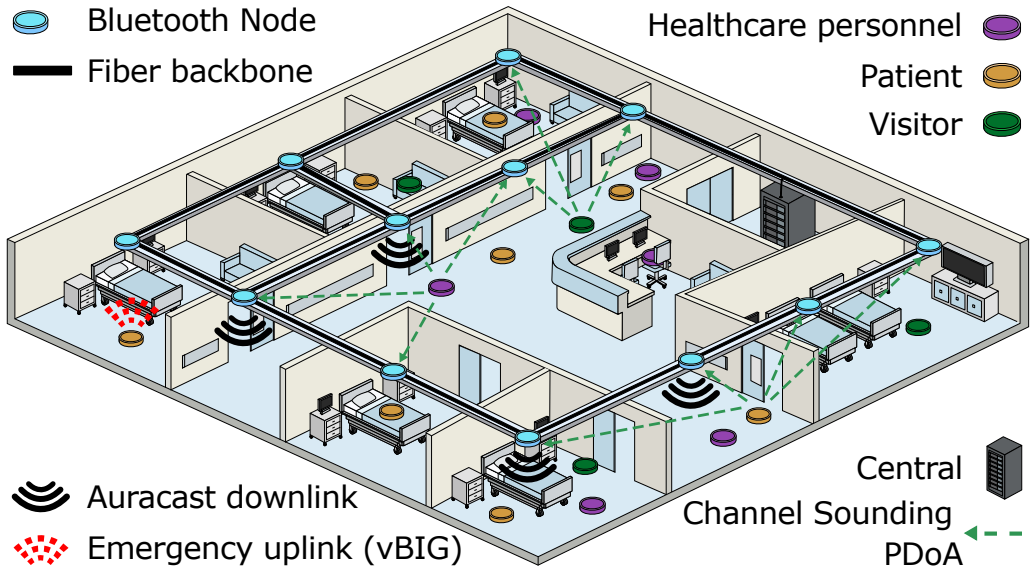


Figure 1: PULSE architecture in a healthcare setting. The infrastructure utilizes a synchronized wired backbone to enable PDoA localization of patients and staff, combined with audio dissemination via Auracast.

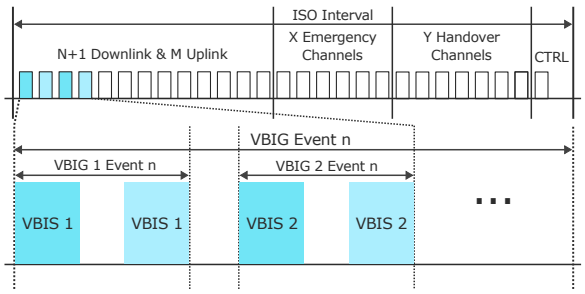


Figure 2: Example of the four stream types for PULSE.

Table 1: PULSE components and their key challenges.

Component	Key challenges
One-way PDoA	Anchor coherence, sub-ns sync, multipath
Unified schedule (vBIG)	Capacity, contention, latency
Low-power tags	Infrastructure dependence
Closed-loop operation	Latency, reliability guarantees
Shared 2.4 GHz medium	Coexistence, trust, privacy

4.1.1 Anchor Coherence. The local-oscillator term $\Delta\phi^{f}_{\mathrm{LO}}$ does not cancel. Each time an anchor retunes to the next tone in the sweep, its frequency synthesizer re-locks at an arbitrary phase, unless the synthesizer is deliberately designed to be phase-deterministic across channels. Two-way CS cancels each device's LO phase through the round trip, a one-way design gives up that mechanism and must instead solve the problem in hardware. An anchor therefore has to either use phase-deterministic synthesis or capture the full 80 MHz band at once with a wideband front end. Neither capability is available in commercial BLE transceivers, although phase-coherent operation across hops has been demonstrated with a dedicated Phase-Locked Loop (PLL) on custom hardware [1]. In PULSE, anchors are thus better understood as custom coherent receivers, while only the tags stay near-commodity.

4.1.2 Backbone Synchronization. The second surviving term is the inter-anchor timing error $\tau_{1,2}$. Because it enters the range difference directly as $c\,\tau_{1,2}$, an error of only 1 ns already corresponds to 30 cm of ranging error. Reaching decimeter accuracy therefore requires the residual jitter between anchors to stay on the order of 300 ps, comparable to the 2.4 GHz carrier period itself (about 416 ps). Precision Time Protocol (PTP) with hardware timestamping typically settles in the tens of nanoseconds, far short of this. The backbone therefore needs White Rabbit-level synchronization or RF reference distribution over fiber.

4.1.3 Multipath and Body Shadowing. Even with ideal anchors and synchronization, indoor propagation remains a dominant source of error. The 80 MHz of available bandwidth resolves multipath components only down to $c/B \approx 3.7$ m, so separating the direct path from its reflections relies on super-resolution techniques that are sensitive to noise and geometry. Hospitals are a particularly complex environment, full of metallic surfaces and constant movement, and a body-worn tag is further attenuated by 15–30 dB whenever the body blocks the line of sight to an anchor. Since hyperbolic positioning amplifies a biased range difference, robust sub-meter accuracy is realistic only under line of sight.

4.2 Unified Schedule

Reusing the vBIG event as a shared schedule keeps coordination simple, but fixes an upper bound on how many tags the network can serve. For example, a 20 ms ISO interval may leave roughly 5 ms per event after audio subevents. This allows one 4 ms tone sweep within

one event, giving on the order of 50 position fixes per second across the network. In a zone with 200 tags, each tag would therefore be updated only once every 4 s. Update rate, audio load, and tag density therefore trade off directly. The emergency channel introduces a separate concern: because it is contention-based, alarms from two patients at the same instant may collide. Repeating each distress burst across several slots and consecutive events rapidly reduces the probability of a sustained collision, keeping worst-case delivery within a few ISO intervals under realistic loads.

4.3 Low-Power Tags

Pushing all coherence and synchronization onto the backbone keeps the tag simple and low power. A vBIG tag synchronizes like an ordinary Auracast receiver and may optionally consume an audio stream, for instance when integrated into a hearing aid. A tag dedicated to ranging and emergency signaling, however, does not need to decode any audio payload. It only stays synchronized to know when to transmit its tones and signal emergencies, avoiding the cost of audio reception. It is otherwise transmit-dominant, with no connection, scanning, or acknowledgments, and removing the handshakes [5] and response slots of prior schemes eliminates further radio activity. What remains is a periodic wake-up to correct crystal drift, which could be made infrequent through a skip mechanism analogous to periodic advertising, though the resulting duty cycle and energy cost remain to be quantified. The trade-off is dependence: the tag is only as reliable as the infrastructure it offloads to, making backbone availability and graceful degradation under partial synchronization loss key concerns.

4.4 Closed-Loop Operation

Closing the loop, from detecting an event to acting on it, introduces its own timing and reliability requirements. A life-safety response such as paging the nearest caregiver after a detected fall must traverse several stages: ranging the patient, reaching a decision at the central unit, and delivering a notification over Auracast. Each stage adds latency and its own failure modes, so the end-to-end path cannot be treated as best-effort. Bounding the total latency, and providing sensible fallback behavior when a stage fails, are necessary for the system to be trusted in a clinical setting, and establishing these bounds is left to future work.

4.5 Shared Medium

Co-locating ranging and audio on one 2.4 GHz broadcast medium raises coexistence and trust concerns. Hospitals are saturated with Wi-Fi and BLE; the vBIG schedule avoids internal collisions but not external interference, which can be mitigated by treating interfered tones as erasures [7] and retransmitting audio, but the resilience of these mechanisms under realistic hospital conditions still needs to be established. Trust is the second concern: because anything broadcast can be overheard, devices must be bootstrapped through Out-Of-Band (OOB) provisioning, for example by scanning a QR code, group and per-device keys are needed to protect the audio and the PDoA data respectively, and resolvable private addresses help keep tags from being tracked by third parties. Continuously localizing patients also raises consent and data-protection questions that computing positions on-premise only partially addresses.

5 Conclusions

PULSE envisions joint tracking and dissemination on a single Bluetooth-derived infrastructure in healthcare environments. Its central insight is that one-way PDoA cancels the tag's transmit phase across synchronized anchors, shifting all coherence requirements to the mains-powered backbone. This keeps wearables simple while Auracast and vBIG provide dissemination and an emergency uplink on the same schedule. Realizing this vision requires phase-coherent custom anchors, sub-nanosecond synchronization, multipath-robust estimation, and protocol behavior beyond today's Bluetooth specifications. We envisage a staged validation process. First, a two-anchor shared-clock software-defined radio setup to verify phase-deterministic capture, calibration, and the jitter-to-error relationship. Next, a multi-anchor deployment would characterize accuracy under realistic conditions. Finally, a vBIG scheduling emulation would assess update rate and emergency latency under audio load.

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